

SYDNEY TECHNICAL HIGH SCHOOL
(Est 1911)

MATHEMATICS EXTENSION II

HSC ASSESSMENT TASK 1

MARCH 2002

Time allowed : 70 minutes

Instructions :

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- This test forms part of your HSC assessment.
- These questions are to be handed in with your answers.

Name : _____

Question 1	Question 2	Question 3	Total

Question 1 (16 marks) - Use a separate sheet of paper

Marks

a) Evaluate i^{2002} .

1

b) If $z = 2 + 2\sqrt{3}i$ find

i) $\frac{z}{1+i}$ in the form $a+ib$.

1

ii) $\bar{z}$.

1

iii) $|z|$

1

iv) $\text{Arg}(z)$

1

v) $\frac{1}{z^4}$ in modulus argument form. (use principle argument)

2

c) On an Argand diagram sketch the locus specified by

$$\text{Arg}(z - 1 + i) = \frac{3\pi}{4}$$

2

d) i) For what values of λ is the equation

2

$$\frac{x^2}{14-\lambda} + \frac{y^2}{\lambda-6} = 1 \quad \text{the locus of an ellipse?}$$

ii) For what values of λ does the equation

1

$$\frac{x^2}{14-\lambda} + \frac{y^2}{\lambda-6} = 1 \quad \text{represent an ellipse with its foci on the } y \text{ axis?}$$

e) Find the volume generated when the area bounded by $x = y^2 - 1$

4

and the y axis is rotated about the line $x = 1$.

Question 2 (16 marks) - Use a separate piece of paper.

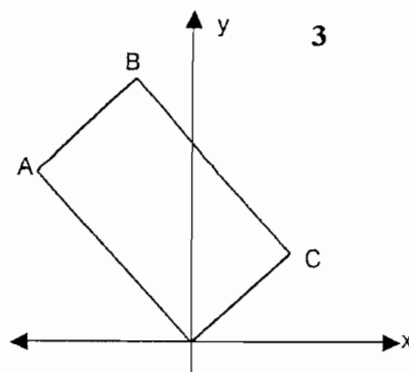
Marks

- a) OABC is a rectangle with $OA = 3 \times OC$.

O is the origin.

If C represents the complex number w , find in terms of w the complex number represented by;

- i) the point A
- ii) the point B
- iii) the vector from A to C



- b) For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$

- Find
- i) the eccentricity.
 - ii) the coordinates of the foci.
 - iii) the equation of the directrices.

- c) i) Show that the equation of the tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$

at the point $P(x_1, y_1)$ is given by $\frac{xx_1}{4^2} + \frac{yy_1}{3^2} = 1$.

- ii) Given that the tangent found in part i) cuts both directrices

of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ above the x axis, find the area

enclosed by this tangent, the two directrices and the x axis.

- d) z is a complex number where $|z| = 1$ and $\text{Arg}(z) = A$.

Find, in term of A the value of $\text{Arg}(z+1)$. Justify your answer.

Question 3 (16 marks) - *Use a separate piece of paper.*

Marks

- a) On an Argand diagram sketch the locus specified by

2

$$|z - 1 - i| > |z|$$

- b) Find the modulus of $\frac{(2-i)^8}{(2+i)^6}$.

3

- c) i) Solve $z^5 = -1$ over the complex field.

2

- ii) If ω is the complex root of $z^5 = -1$ with smallest

3

positive argument, show that the other complex roots equal

$$-\omega^2, \omega^3 \text{ and } -\omega^4.$$

- iii) Using ω as in part ii) simplify $(1 - \omega + \omega^2 - \omega^3)^8$

3

- d) Describe and sketch on an Argand diagram the locus

3

of z given that $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$ where $z \neq 0$.

SOLUTIONS

QUESTION 1

$$\begin{aligned} \text{a. } i^{2002} &= (i^2)^{1001} \\ &= (-1)^{1001} \\ &= -1 \end{aligned}$$

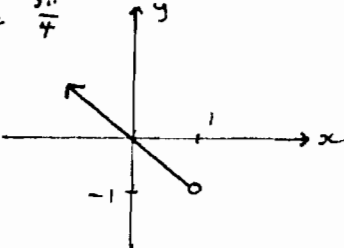
$$\begin{aligned} \text{b. i)} \quad \frac{2+2\sqrt{3}i}{1+i} &= \frac{2+2\sqrt{3}i}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{2-2i+2\sqrt{3}i+2\sqrt{3}}{2} \\ &= (1+\sqrt{3}) + i(\sqrt{3}-1) \end{aligned}$$

$$\text{ii)} \quad \bar{z} = 2 - 2\sqrt{3}i$$

$$\begin{aligned} \text{iii)} \quad |z| &= \sqrt{2^2 + (2\sqrt{3})^2} \\ &= 4 \end{aligned}$$

$$\text{iv)} \quad \text{Arg}(z) = \frac{\pi}{3}$$

$$\begin{aligned} \text{v)} \quad \frac{1}{z^4} &= z^{-4} \\ &= [4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})]^{-4} \\ &= 4^{-4} (\cos \frac{-4\pi}{3} + i \sin \frac{-4\pi}{3}) \\ &= \frac{1}{256} (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) \end{aligned}$$

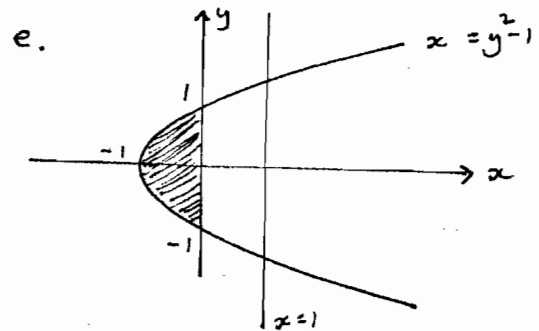
$$\text{c. } \text{Arg}(z - (1-i)) = \frac{3\pi}{4}$$


$$\begin{aligned} \text{d. i)} \quad 14 - \lambda > 0 \text{ and } \lambda - 6 > 0 \\ \lambda < 14 \text{ and } \lambda > 6 \end{aligned}$$

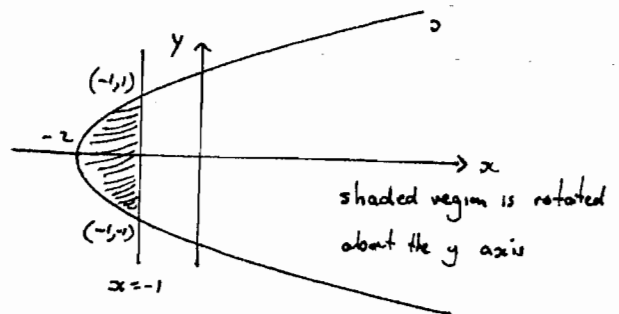
$$\therefore 6 < \lambda < 14, \lambda \neq 10$$

$$\begin{aligned} \text{ii)} \quad \lambda - 6 > 14 - \lambda \text{ (plus part i)} \\ \lambda > 10 \end{aligned}$$

$$\therefore 10 < \lambda < 14$$



translate 1 unit left



$$\begin{aligned} \therefore V &= 2\pi \int_0^1 (y^2 - 2)^2 - (-1)^2 dy \\ &= 2\pi \int_0^1 y^4 - 4y^2 + 3 dy \\ &= 2\pi \left[\frac{1}{5} y^5 - \frac{4}{3} y^3 + 3y \right]_0^1 \\ &= 2\pi \left[\frac{1}{5} - \frac{4}{3} + 3 \right] \\ &= \frac{56\pi}{15} \text{ cu units} \end{aligned}$$

Question 2

a i) $3iw$

ii) $3iw + w$
or $w(1+3i)$

iii) $w - 3iw$
or $w(1-3i)$

b. i) $b^2 = a^2(1-e^2)$
 $9 = 16(1-e^2)$

$e = \frac{\sqrt{7}}{4}$

ii) foci $(\pm ae, 0)$
 $(\pm\sqrt{7}, 0)$

iii) directrices $x = \pm \frac{a}{e}$

$x = \pm \frac{16}{\sqrt{7}}$

c. differentiating

$\frac{2x}{16} + \frac{2y}{9} \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = \frac{-9x}{16y}$

$\therefore$ gradient of tangent at $P(x_1, y_1)$ is

$m_T = \frac{-9x_1}{16y_1}$

$\therefore$ equation of tangent

$y - y_1 = m(x - x_1)$

$y - y_1 = \frac{-9x_1}{16y_1}(x - x_1)$

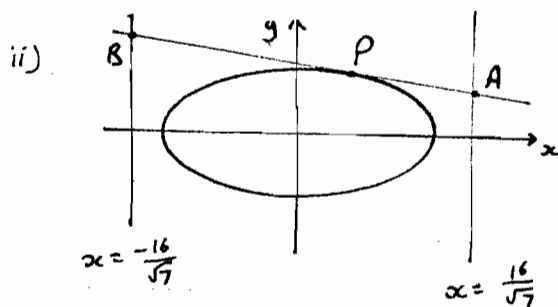
$16yy_1 - 16y_1^2 = -9xx_1 + 9x_1^2$

$9xx_1 + 16yy_1 = 9x_1^2 + 16y_1^2$ ($\div 144$)

$\frac{xx_1}{16} + \frac{yy_1}{9} = \frac{x_1^2}{16} + \frac{y_1^2}{9}$

but (x_1, y_1) lies on $\frac{x^2}{16} + \frac{y^2}{9} = 1$

$\therefore \frac{xx_1}{16} + \frac{yy_1}{9} = 1$



to find A: sub $x = \frac{16}{\sqrt{7}}$ into tangent

$\therefore \frac{\frac{16}{\sqrt{7}}x_1}{16} + \frac{yy_1}{9} = 1$

$y = \frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}}\right)$

$\therefore A \left(\frac{16}{\sqrt{7}}, \frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}}\right) \right)$

to find B: sub $x = \frac{-16}{\sqrt{7}}$ into tangent

$\frac{\frac{-16}{\sqrt{7}}x_1}{16} + \frac{yy_1}{9} = 1$

$y = \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}}\right)$

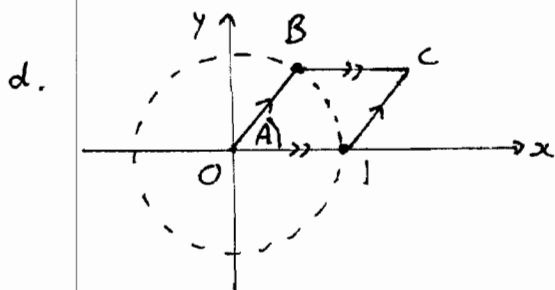
$\therefore B \left(\frac{-16}{\sqrt{7}}, \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}}\right) \right)$

$$\therefore \text{Area} = \frac{b}{2} (a+b)$$

$$= \frac{1}{2} \cdot \frac{32}{\sqrt{7}} \left(\frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}} \right) + \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}} \right) \right)$$

$$= \frac{144}{\sqrt{7} y_1} (1+1)$$

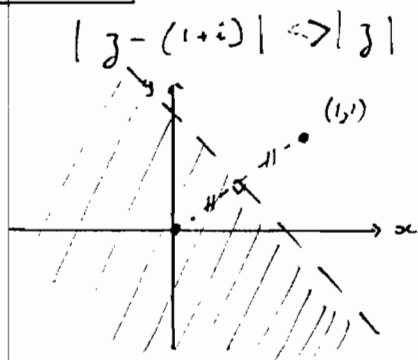
$$= \frac{288}{\sqrt{7} y_1} \text{ sq units}$$



B represents z
 C represents $z+1$
 shape is a rhombus
 diagonals bisect angles

$$\therefore \text{Arg}(z+1) = \frac{A}{2}$$

QUESTION 3



b. $|2-i| = \sqrt{5}, |2+i| = \sqrt{5}$

$$\therefore \left| \frac{(2-i)^8}{(2+i)^6} \right| = \frac{\sqrt{5}^8}{\sqrt{5}^6} = 5$$

c. i) roots of $z^5 = -1$ are

$$z_1 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$z_2 = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

$$z_3 = -1$$

$$z_4 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}$$

$$z_5 = \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}$$

ii) $\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = w$

$$\begin{aligned} \therefore \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^3 \\ &= w^3 \end{aligned}$$

$$\begin{aligned} \therefore \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^7 \\ &= w^7 \\ &= w^5 \cdot w^2 \quad (w^5 = -1) \\ &= -w^2 \end{aligned}$$

$$\begin{aligned} \therefore \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^9 \\ &= w^9 \\ &= w^5 \cdot w^4 \\ &= -w^4 \end{aligned}$$

iii) sum of roots of $z^5 + 1 = 0$ equals 0

$$\begin{aligned} \therefore -1 + w - w^2 + w^3 - w^4 &= 0 \\ \text{or } 1 - w + w^2 - w^3 &= -w^4 \end{aligned}$$

$$\therefore (1 - w + w^2 - w^3)^8 = (-w^4)^8$$

$$\begin{aligned}
 &= w^{32} \\
 &= w^{30} \cdot w^2 \\
 &= (w^5)^6 \cdot w^2 \\
 &= (-1)^6 \cdot w^2 \\
 &= w^2 \\
 &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}
 \end{aligned}$$

d. $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0 \quad z \neq 0$

$$\operatorname{Re}\left(x+iy - \frac{1}{x+iy}\right) = 0$$

$$\operatorname{Re}\left(x+iy - \frac{1}{x+iy} + \frac{x-iy}{x-iy}\right) = 0$$

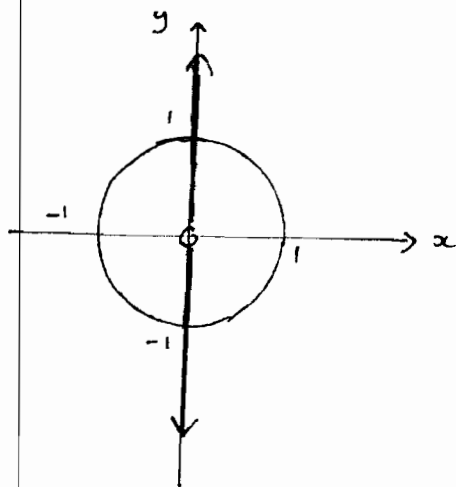
$$\operatorname{Re}\left(x+iy - \frac{x-iy}{x^2+y^2}\right) = 0$$

$$\therefore x - \frac{x}{x^2+y^2} = 0$$

$$x(x^2+y^2) - x = 0$$

$$x(x^2+y^2-1) = 0$$

$$\therefore x=0 \text{ or } x^2+y^2=1 \quad (z \neq 0)$$



QUESTION 1

$$\begin{aligned}
 \text{a. } i^{2002} &= (i^2)^{1001} \\
 &= (-1)^{1001} \\
 &= -1
 \end{aligned}$$

1 mark

$$\begin{aligned}
 \text{b. i)} \quad \frac{2+2\sqrt{3}i}{1+i} &= \frac{2+2\sqrt{3}i}{1+i} \times \frac{1-i}{1-i} \\
 &= \frac{2-2i+2\sqrt{3}i+2\sqrt{3}}{2} \\
 &= (1+\sqrt{3}) + i(\sqrt{3}-1)
 \end{aligned}$$

1 mark

$$\text{ii)} \quad \bar{z} = 2 - 2\sqrt{3}i$$

1 mark

$$\begin{aligned}
 \text{iii)} \quad |z| &= \sqrt{2^2 + (2\sqrt{3})^2} \\
 &= 4
 \end{aligned}$$

1 mark

$$\text{iv)} \quad \text{Arg}(z) = \frac{\pi}{3}$$

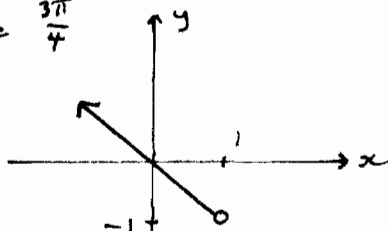
1 mark

$$\begin{aligned}
 \text{v)} \quad \frac{1}{z^4} &= z^{-4} \\
 &= \left[4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \right]^{-4} \\
 &= 4^{-4} \left(\cos -\frac{4\pi}{3} + i \sin -\frac{4\pi}{3} \right) \\
 &= \frac{1}{256} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)
 \end{aligned}$$

must have argument $\frac{2\pi}{3}$

2 marks

$$\text{c. } \text{Arg}(z - (1-i)) = \frac{3\pi}{4}$$

must go through origin
must have open circle

2 marks

d. i) $14 - \lambda > 0$ and $\lambda - 6 > 0$
 $\lambda < 14$ and $\lambda > 6$

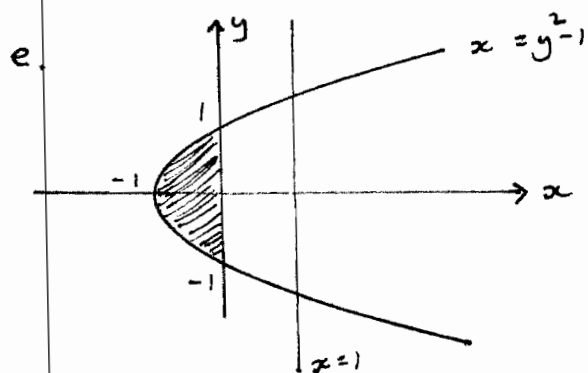
$\therefore 6 < \lambda < 14, \lambda \neq 10$

ii) $\lambda - 6 > 14 - \lambda$ (plus part i)
 $\lambda > 10$

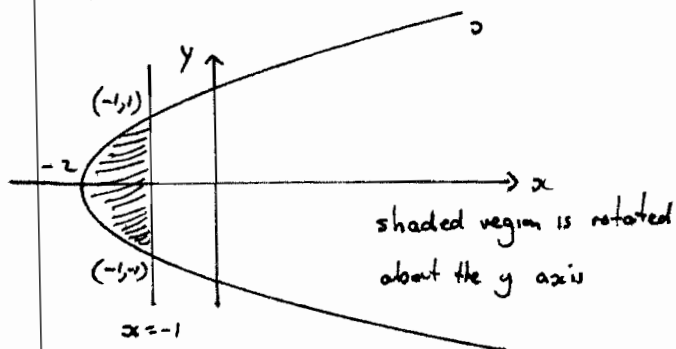
$\therefore 10 < \lambda < 14$

2 marks

1 mark



translate 1 unit left



$\therefore V = 2\pi \int_0^1 (y^2 - 2)^2 - (-1)^2 dy$
 $= 2\pi \int_0^1 y^4 - 4y^2 + 3 dy$
 $= 2\pi \left[\frac{1}{5} y^5 - \frac{4}{3} y^3 + 3y \right]_0^1$
 $= 2\pi \left[\frac{1}{5} - \frac{4}{3} + 3 \right]$
 $= \frac{56\pi}{15} \text{ cu units}$

Can use any other method

4 marks

QUESTION 2

a i) $3iw$

1 mark

ii) $3iw + w$
or $w(1+3i)$

1 mark

iii) $w - 3iw$
or $w(1-3i)$

1 mark

b. i) $b^2 = a^2(1-e^2)$
 $9 = 16(1-e^2)$

$$e = \frac{\sqrt{7}}{4}$$

1 mark

ii) foci $(\pm ae, 0)$
 $(\pm\sqrt{7}, 0)$

1 mark

iii) directrices $x = \pm \frac{a}{e}$

$$x = \pm \frac{16}{\sqrt{7}}$$

1 mark

c. differentiating

$$\frac{2x}{16} + \frac{2y \frac{dy}{dx}}{9} = 0$$

$$\therefore \frac{dy}{dx} = \frac{-9x}{16y}$$

$\therefore$ gradient of tangent at $P(x_1, y_1)$ is

$$m_T = \frac{-9x_1}{16y_1}$$

$\therefore$ equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = \frac{-9x_1}{16y_1} (x - x_1)$$

$$16yy_1 - 16y_1^2 = -9xx_1 + 9x_1^2$$

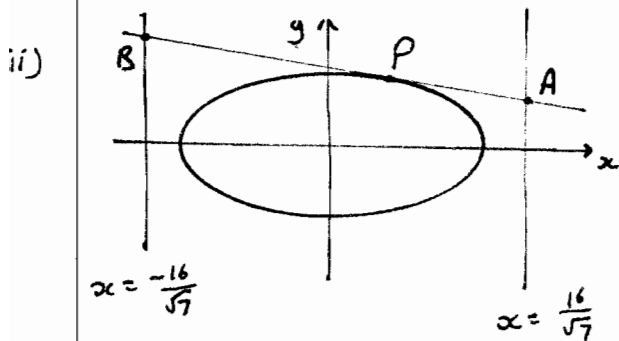
$$9xx_1 + 16yy_1 = 9x_1^2 + 16y_1^2 \quad (\div 144)$$

$$\frac{xx_1}{16} + \frac{yy_1}{9} = \frac{x_1^2}{16} + \frac{y_1^2}{9}$$

$$\text{but } (x_1, y_1) \text{ lies on } \frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\therefore \frac{xx_1}{16} + \frac{yy_1}{9} = 1$$

4 marks



to find A: sub $x = \frac{16}{\sqrt{7}}$ into tangent

$$\therefore \frac{\frac{16}{\sqrt{7}} x_1}{16} + \frac{yy_1}{9} = 1$$

$$y = \frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}} \right)$$

$$\therefore A \left(\frac{16}{\sqrt{7}}, \frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}} \right) \right)$$

to find B: sub $x = -\frac{16}{\sqrt{7}}$ into tangent

$$\frac{-\frac{16}{\sqrt{7}} x_1}{16} + \frac{yy_1}{9} = 1$$

$$y = \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}} \right)$$

$$\therefore B \left(-\frac{16}{\sqrt{7}}, \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}} \right) \right)$$

$$\therefore \text{Area} = \frac{h}{2} (a+b)$$

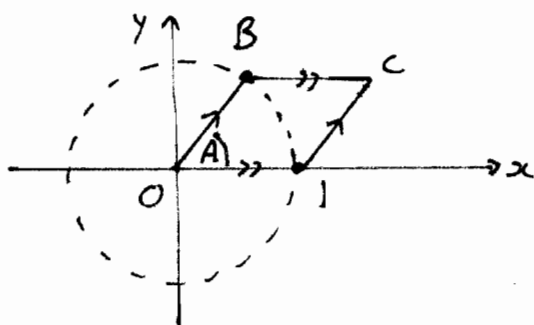
$$= \frac{1}{2} \cdot \frac{32}{\sqrt{7}} \left(\frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}} \right) + \frac{9}{y_1} \left(1 + \frac{x_1}{\sqrt{7}} \right) \right)$$

$$= \frac{144}{\sqrt{7} y_1} (1+1)$$

$$= \frac{288}{\sqrt{7} y_1} \text{ sq units}$$

4 marks

d.



B represents z
 C represents $z+1$
 shape is a rhombus
 diagonals bisect angles

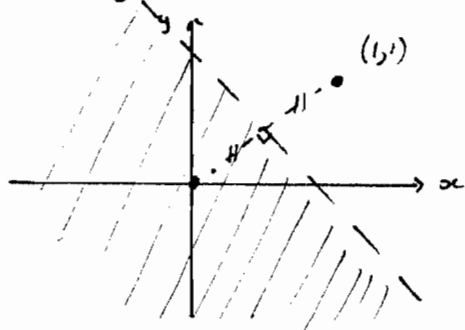
can be done other ways
 need some attempt at
 justifying answer.

$$\therefore \text{Arg}(z+1) = \frac{A}{2}$$

2 marks

Question 3

a. $|z - (1+i)| > |z|$



2 marks

b. $|2-i| = \sqrt{5}, |2+i| = \sqrt{5}$

$$\therefore \left| \frac{(2-i)^8}{(2+i)^6} \right| = \frac{\sqrt{5}^8}{\sqrt{5}^6}$$

$$= 5$$

3 marks

c. i) roots of $z^5 = -1$ are

$$z_1 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$z_2 = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

$$z_3 = -1$$

$$z_4 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}$$

$$z_5 = \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}$$

roots can be in
any form

2 marks

ii) $\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = w$

$$\begin{aligned} \therefore \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^3 \\ &= w^3 \end{aligned}$$

$$\begin{aligned} \therefore \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^7 \\ &= w^7 \\ &= w^5 \cdot w^2 \quad (w^5 = -1) \\ &= -w^2 \end{aligned}$$

$$\begin{aligned} \therefore \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} &= \\ &= (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^9 \\ &= w^9 \\ &= w^5 \cdot w^4 \\ &= -w^4 \end{aligned}$$

3 marks

iii) sum of roots of $z^5 + 1 = 0$ equals 0

$$\therefore -1 + w - w^2 + w^3 - w^4 = 0$$

$$\text{or } 1 - w + w^2 - w^3 = -w^4$$

$$\therefore (1 - w + w^2 - w^3)^8 = (-w^4)^8$$

$$\begin{aligned}
 &= w^{32} \\
 &= w^{30} \cdot w^2 \\
 &= (w^5)^6 \cdot w^2 \\
 &= (-1)^6 \cdot w^2 \\
 &= w^2 \\
 &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}
 \end{aligned}$$

give full marks for w^2

3 marks.

d. $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0 \quad z \neq 0$

$$\operatorname{Re}\left(x+iy - \frac{1}{x+iy}\right) = 0$$

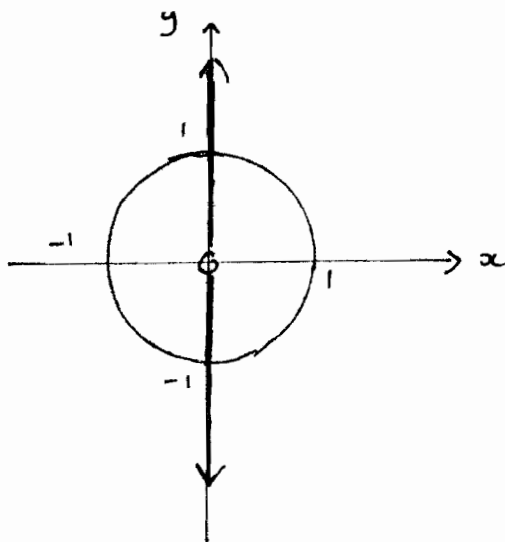
$$\operatorname{Re}\left(x+iy - \frac{1}{x+iy} + \frac{x-iy}{x-iy}\right) = 0$$

$$\operatorname{Re}\left(x+iy - \frac{x-iy}{x^2+y^2}\right) = 0$$

$$\therefore x - \frac{x}{x^2+y^2} = 0$$

$$\begin{aligned}
 x(x^2+y^2) - x &= 0 \\
 x(x^2+y^2-1) &= 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore x=0 \text{ or } x^2+y^2=1 \\
 (z \neq 0)
 \end{aligned}$$



origin needs to be excluded.

3 marks